A Century Is Not Enough: Certification Bounds for Out-of-Sample Return Predictability

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Abstract

How much predictability could a century of data rule out? We answer in closed form and then measure it. Evidence about an alpha accumulates as log wealth, which grows at half the squared information ratio, so a sample spanning T years can certify an annualised information ratio no smaller than a floor that depends on T, the level, and the size of the search—and on nothing else. Sampling the same span more finely does not lower it: the floor for the Goyal–Welch data is 0.38 monthly, 0.37 quarterly and 0.35 annually, across a twelvefold range in observation count. Against the effect Campbell and Thompson call economically significant, an annualised ratio of 0.25, a single predictor needs 99 years to reach 80% power and the 46-predictor search of Goyal et al. [1] needs 244. The canonical dataset is 100 years old. We then apply an anytime-valid certificate to 117 predictor–frequency pairs spanning 1926–2025. Nothing is certified, under any of nine specifications; the largest evidence value ever observed is 8.0 against the 20 required. The deliverable is not that null but the bound beside it: the median predictor's incremental annualised information ratio is at most 0.60, and 112 of 116 ceilings lie above 0.25, so for almost every published predictor a century of the best-studied data in finance can separate neither "nothing" nor "economically meaningful" from the other. Because evidence compounds, the wealth path also dates each edge: the valuation ratios d/p, d/y and e/p—the predictors a Clark–West test ranks highest—stopped contributing in the 1950s and have added nothing since, which a static test cannot see. Our pipeline reproduces Goyal et al.'s published out-of-sample R^2_{OS} with a correlation of 0.968 across their fourteen classic predictors.

1 Introduction

The out-of-sample predictability literature asks whether a variable forecasts the equity premium. After Welch and Goyal [2] the consensus answer is mostly no, and after Goyal et al. [1] it is mostly no again for a further twenty-nine candidates. This paper asks a prior question that determines what any such answer can mean: *given a sample of a certain length, what size of effect could it have detected at all?*

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Preprint. Code, pinned data, and one script per reported number: https://github.com/ITheClixs/crypto-return-predictability

The question has an exact answer, and the answer is uncomfortable. Section 2 derives a certification floor: the smallest annualised information ratio a sample spanning T years can establish at level α. Two features of it matter. The floor scales with $1/\sqrt{T}$ in *calendar span* and is almost unaffected by sampling frequency, so a century of monthly data and a century of annual data face nearly the same bound despite a twelvefold difference in observations. And it degrades with the size of the search, which in this literature is large and known.

Put those together against the effect the literature itself treats as meaningful—Campbell and Thompson [3] take a monthly out-of-sample R^2_{OS} of 0.5%, an annualised information ratio of about 0.25, as economically significant—and the arithmetic is stark. A single predictor needs 99 years to reach 80% power against it. Correcting for the 46-predictor search that Goyal et al. [1] actually ran needs 244. The canonical dataset begins in 1926.

What we do with that. A bound on what a study could have found is only useful if paired with a bound on what it did find. Section 3 uses an anytime-valid *certificate* [4], a non-negative wealth process whose validity is a finite-sample theorem and whose logarithm is the profit and loss of an explicit drift-hedged strategy. Two properties earn its place here. It supplies a time-uniform *ceiling* on a predictor's incremental value, which is the number this literature has never reported and the one a reader can act on. And because evidence compounds rather than averages, its wealth path *dates* an edge: a predictor that worked until 1957 and not since cannot keep accumulating, and the path shows exactly when it stopped.

What we find. Across 117 predictor–frequency pairs from 1926 to 2025, nothing is certified, under any of nine specifications of burn-in, window and payoff. That is unsurprising and is not the contribution. The contribution is the interval: the median predictor's incremental annualised information ratio is bounded above by 0.60, and 112 of 116 ceilings sit above Campbell and Thompson's 0.25. A century of the most heavily studied data in finance can rule out neither zero nor an economically meaningful edge, for almost every predictor in the set.

The dating result is the one we did not expect. Clark–West and the certificate rank the predictors close to inversely, and the wealth paths say why. The valuation ratios d/p, d/y and e/p—the most cited predictors in the literature, and the ones Clark–West scores highest at 1.34 to 1.64—peaked in 1946–1957 and have contributed nothing in the seventy years since. Inflation and the Treasury-bill rate, which Clark–West scores lowest, are the only variables still accumulating. A test that averages a covariance over a century cannot distinguish a steady edge from a dead one; a test that compounds cannot avoid it.

What this is not. We do not claim the certificate is more powerful than Clark–West. It is not: Guven [4] measures the exchange rate at roughly a factor of two in sample size, the price of validity at every stopping time and under weaker assumptions. Nor do we claim to overturn Welch and Goyal [2]; we agree with it and reproduce it. The claim is narrower and, we think, more useful: this literature has been operating at or below the detection threshold of its own data for its entire history, the threshold is computable in advance, and the quantity worth reporting is a ceiling rather than a p-value.

2 The certification bound

Let a signal-weighted strategy have annualised information ratio IR, and consider testing whether $\mathrm{IR} > 0$ from T years of data at level α.

Proposition 1 (Certification floor). *A one-sided fixed-sample test at level α attains power β against an alternative* IR *if and only if $T \geq (z_\alpha + z_\beta)^2/\mathrm{IR}^2$ years. Equivalently, a sample spanning T years cannot establish any ratio below*

$$\mathrm{IR}_{\min}(T, \alpha, \beta) \;=\; \frac{z_\alpha + z_\beta}{\sqrt{T}}.$$

Under a Bonferroni correction for a search over m candidates, replace α by α/m.

The proposition is elementary and that is the point: it needs no new theory and it applies to the tools this literature already uses. Its content is in the substitutions.

Table 1: Years of data needed to reach 80% power against the effect Campbell and Thompson [3] call economically significant—a monthly out-of-sample R^2_{OS} of 0.5%, an annualised information ratio of 0.25—as the search widens. The searches are those the literature actually ran. The Goyal–Welch sample begins in 1926.

Correction	Floor at 100 years	Years needed for IR $= 0.25$
One predictor, $\alpha = 0.05$	0.252	99
Bonferroni over 17 [2]	0.365	207
Bonferroni over 46 [1]	0.397	244
Bonferroni over 100	0.420	273
Available: 100 years (1926–2025).		

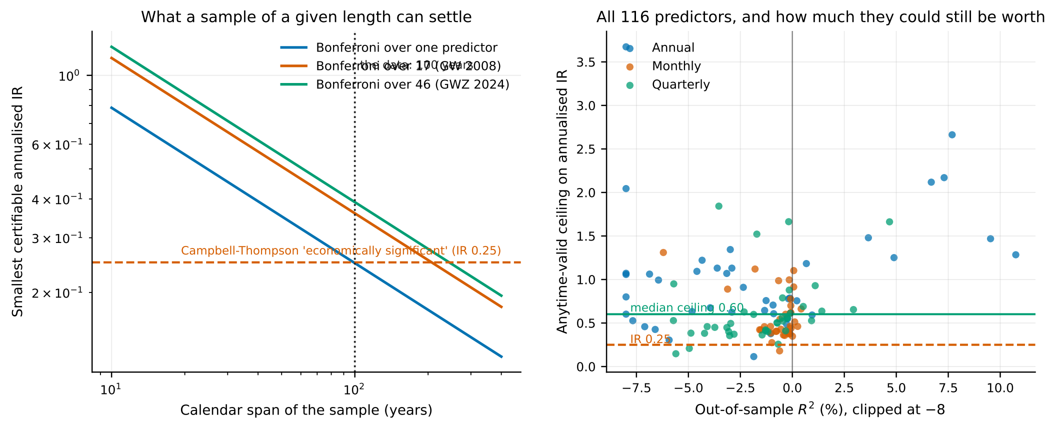

Figure 1: **Left:** the certification floor against calendar span, for searches of one, seventeen and forty-six predictors. The horizontal line is Campbell and Thompson's economic-significance threshold; the vertical line is the data. **Right:** every predictor–frequency pair, plotted as its anytime-valid ceiling against its out-of-sample R^2_{OS}. The median ceiling is 0.60 and 112 of 116 lie above 0.25: for almost every predictor the data rule out neither zero nor an economically meaningful edge.

Corollary 2 (Frequency does not help)**.** $\mathrm{IR}_{\min}$ *depends on the calendar span T and not on the sampling frequency. Increasing the number of observations by sampling a fixed span more finely leaves the floor unchanged.*

This is worth stating because the opposite is widely assumed. Doubling the observation count by moving from quarterly to monthly data halves the standard error of a *per-period* mean, but the annualised information ratio is a per-period quantity scaled by $\sqrt{P}$, and the two effects cancel exactly. Section 5 confirms it on the data: the measured floor is 0.38, 0.37 and 0.35 at monthly, quarterly and annual frequency over the same century.

Table 1 and the left panel of Figure 1 state the position. A single predictor tested once sits within one year of the boundary, with no margin for parameter instability, transaction costs, or a slightly smaller true effect. Corrected for the search that was actually performed, the literature is below the line by a factor of two to three in required span.

3 The instrument

We use the drift-robust certificate of Guven [4] and summarise only what is needed here. For a predictable signal g_t—the model's forecast minus the intercept-only forecast from the same estimation window—and an outcome y_t, the certificate is

$$\mathcal{E}_t \;=\; \inf_{\mu\in\mathbb{R}} \Big[w\, W^{\mathrm{sig}}_t(\mu) + (1-w)\, W^{\mathrm{ctr}}_t(\mu) \Big],$$

where W^{sig} is the wealth of a bettor staking a predictable fraction on g_t leading y_t, and W^{ctr} is the wealth of a bettor staking two-sidedly on μ being the wrong drift. Each is a non-negative martingale for fixed μ, so their convex combination is one, and an infimum is at most its value at the truth; Ville's inequality then gives $P(\exists t : \mathcal{E}_t \geq 1/\alpha) \leq \alpha$ in finite samples and uniformly in time, for any drift. We use the identity payoff throughout, which assumes only a martingale-difference null and an a-priori envelope on $|y_t|$, and report the bounded payoffs as sensitivities.

Three properties are load-bearing here.

A ceiling, not just a verdict. Inverting the same construction over the value parameter gives a time-uniform confidence interval for $\theta = \mathbb{E}[z_t(y_t - \mu)]$, the expected return of a position sized by the standardised signal. Read as an annualised information ratio, its upper endpoint is the number we report.

Evidence compounds, so it can be dated. $\mathcal{E}_t$ is a product. A predictor whose edge stops contributing stops the process, and the path records when.

Multiplicity without a bootstrap. E-values average under arbitrary dependence, so the grid-level statistic over 117 dependent tests is an average, and false-discovery control follows from e-BH [5].

4 Data and design

We use the Goyal et al. [1] dataset through December 2025: monthly, quarterly and annual equity-premium series from 1926 together with the seventeen classic Welch and Goyal [2] predictors and the twenty-nine added by Goyal et al.. The outcome is the log equity premium $\log(1+r_t) - \log(1+r_t^f)$; predictors are lagged one period; forecasts come from an expanding-window OLS regression with an intercept, re-estimated every period, and the benchmark is the same window's historical mean, which is that regression's intercept-only restriction. Following Goyal et al., the out-of-sample period begins twenty years after the in-sample period starts.

Real-time versus revised series. Many of the new predictors require re-estimation every period, and Goyal et al. ship them as vintage matrices whose rows index the period and whose columns index the vintage. The admissible series is the diagonal—what a forecaster actually had—and the revised series is the last column. We use the diagonal throughout and record the source per predictor. On the twenty-five predictors where both exist, the revised series looks *better* for only four, so this choice does not drive the results; it is nonetheless the difference between an out-of-sample claim and a claim that quietly uses full-sample parameter estimates.

Validation. Because everything here depends on the pipeline being right, we validate it against Goyal et al.'s published Table 3 Panel A, predictor by predictor, matching their evaluation window and their Campbell–Thompson restriction. Across the fourteen classic monthly predictors the correlation is 0.968 (Spearman 0.959), signs agree in 13 of 14, and the median absolute gap is 0.075 percentage points. The single disagreement is tms, where both values round to zero.

Remark 3 (The Campbell–Thompson restriction is not free)**.** Truncating the forecast at a non-negative equity premium helps 13 of the 14 predictors and moves the mean out-of-sample R^2_{OS} from -0.511% to -0.281%. Since the effects under discussion are themselves a few tenths of a percentage point, an economically motivated truncation is doing about as much work as any predictor in the set. It helps most exactly where the unrestricted regression produces the wildest forecast. We report unrestricted numbers by default.

5 Results

Nothing is certified, and not for want of trying. Across 117 predictor–frequency pairs, out-of-sample R^2_{OS} is negative for 95 (81%), no predictor reaches the $\alpha = 0.05$ threshold, the grid-level e-value is 1.79 against the 20 required, and nothing survives e-BH. Table 2 sweeps the specification: burn-ins of ten to forty years, rolling windows of twenty to fifty years against the expanding one, and two alternative payoffs. Nothing certifies under any of the nine, and the largest e-value ever observed is 8.04. The rolling windows are the informative negative—Welch and Goyal's complaint is instability, so a window that forgets is the specification most likely to find something, and it does not.

Table 2: Specification sensitivity, fourteen classic monthly predictors. Rejecting needs an e-value of 20.

Burn-in	Window	Payoff	Median R^2_{OS} (%)	Max e-value	Grid e-value	Certified
10 y	expanding	identity	-0.329	3.05	1.48	0
20 y	expanding	identity	-0.378	5.71	2.04	0
30 y	expanding	identity	-0.342	8.02	2.55	0
40 y	expanding	identity	-0.308	6.27	1.94	0
20 y	rolling 20 y	identity	-0.989	5.50	1.81	0
20 y	rolling 30 y	identity	-0.579	8.04	2.73	0
20 y	rolling 50 y	identity	-0.515	7.58	2.17	0
20 y	expanding	tanh	-0.378	1.74	1.24	0
20 y	expanding	sign	-0.378	1.38	1.08	0

Table 3: Clark–West against the certificate, and what the wealth path shows. "Nats" are the log-wealth contributed in each half of the evaluation period.

Predictor	Clark–West	e-value	Peak year	Nats, 1st half	Nats, 2nd half
d/y	1.64	1.14	1957	$+0.13$	$+0.00$
e/p	1.55	1.00	1946	$+0.00$	$+0.00$
tbl	1.44	3.48	2021	$+0.20$	$+1.05$
d/p	1.34	1.45	1957	$+0.37$	$+0.00$
tms	0.78	3.92	1987	$+1.34$	$+0.03$
$infl$	0.83	5.71	1999	$+0.59$	$+1.16$

The ceilings are the result. The right panel of Figure 1 plots every pair. The median ceiling on the incremental annualised information ratio is 0.60; 112 of 116 exceed 0.25. By frequency the medians are 0.46 monthly, 0.53 quarterly and 0.86 annual. The correct summary of a century of this literature is therefore not "the predictors do not work" but a statement with a number in it: *the incremental information ratio of the median canonical predictor lies somewhere between zero and approximately* 0.6*, and a hundred years cannot narrow it further*.

Frequency does not buy power. The certifiable floor is 0.38 monthly, 0.37 quarterly and 0.35 annually over the same span, across 1200, 400 and 100 observations respectively—Corollary 2, measured.

When each edge existed. Table 3 and Figure 2 give the result we did not anticipate. Clark–West averages a covariance over the century; the certificate compounds, so a dead edge stops the process and the path dates the death. The three valuation ratios have added nothing since the 1950s. The two variables still accumulating are the two Clark–West likes least.

One degenerate fit, flagged rather than trusted. The variance risk premium at quarterly frequency reports $R^2_{OS} = -98\%$ alongside an e-value of 4.76. This is not a contradiction: its regression forecasts a -78% quarterly equity premium against an outcome standard deviation of 8%, a nine-sigma artifact of a 120-observation fit. Squared-error measures are quadratic and one such point destroys them; the certificate's stake is capped relative to the signal's own scale and is unmoved. We record the largest forecast each fit ever made in units of the outcome's standard deviation and flag anything beyond five. One of 117 is flagged; excluding it changes nothing.

6 Discussion

A null result and a bound are different objects. Welch and Goyal [2] established that these predictors do not beat the historical mean out of sample, and we reproduce it. What that literature has not supplied is the other half: how large an effect the data could have excluded. Without it a negative result is not interpretable, because it is consistent both with there being nothing and with there being something the design could never have seen. Table 1 shows this literature has lived in the second case throughout.

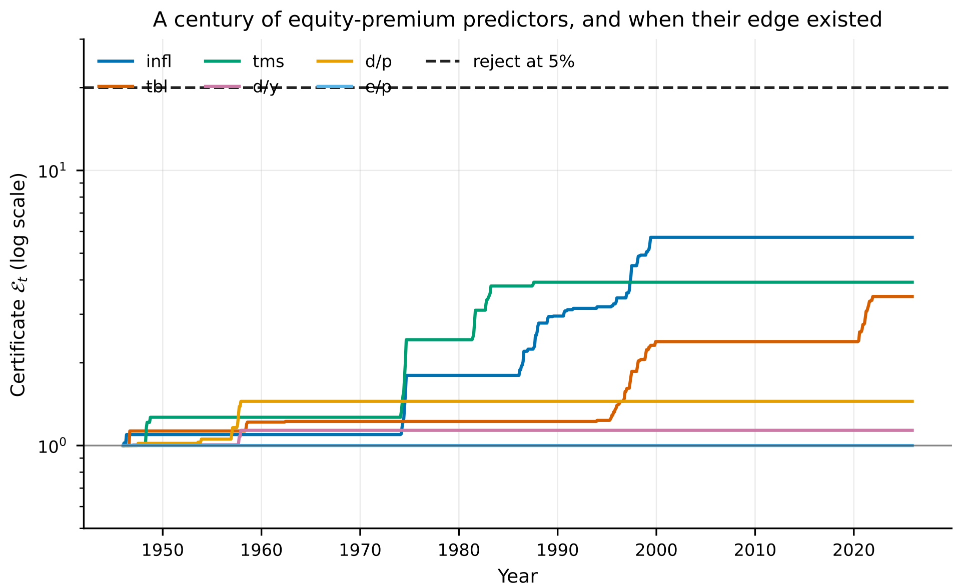

Figure 2: Certificate wealth paths for six classic predictors. A predictor whose edge stops contributing stops compounding, so the path dates it. The valuation ratios flatten in the 1950s; inflation and the Treasury-bill rate are the only two still rising.

The instrument is not the finding. We are explicit that the certificate is less powerful than Clark–West where Clark–West is well specified [4]. It is used here for three things a t-statistic cannot do: produce a ceiling, date an edge, and correct a dependent grid without a bootstrap. Its failure to certify is therefore weak evidence on its own, and the ceilings—which do not depend on rejecting anything—carry the paper.

What would change the answer. Not a better test: Proposition 1 binds any procedure. Not finer sampling: Corollary 2. Only a longer span, a larger effect, or a smaller search. Of those, only the last is under a researcher's control, which is an argument for pre-registering a small number of candidates rather than screening many.

7 Limitations

The floor of Proposition 1 assumes a constant effect; a predictor that is strong in some regimes and absent in others has a higher effective floor still, so our numbers are optimistic. Bonferroni is conservative, and a sharper multiplicity correction would lower the corrected column of Table 1, though not to the point of clearing the boundary. The certificate's own conservatism means its ceilings are wider than a fixed-sample interval would be, which cuts against us. And our predictor set is Goyal et al.'s: a set assembled from published papers is selected on past significance, which biases toward finding predictability, not away from it.

8 Conclusion

Evidence about an alpha accumulates as log wealth, so what a sample can certify is fixed in advance by its calendar span, its level, and the size of the search. For the canonical equity-premium data those quantities give 99 years for a single predictor against the effect the literature calls economically significant, and 244 once the search that was actually run is accounted for. There are 100 years. Applying an anytime-valid certificate to 117 predictor–frequency pairs certifies nothing under any of nine specifications, but the useful output is the interval it does supply: the median predictor's incremental annualised information ratio is at most 0.60 and cannot be separated from zero. A century of the best-studied data in finance is not enough to tell the difference between no predictability and twice as much as would matter—and, because evidence compounds, the same instrument shows that the most cited predictors in the literature stopped contributing seventy years ago.

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